

PKCERT AI & Software Development Internship

Task 07 – Machine Learning Foundations:

Train/Test Split, Model Fit, and Linear Regression

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1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed. It is widely used in real-world applications such as price prediction, recommendation systems, fraud detection, and medical diagnosis. In this task, the fundamental concepts of machine learning were explored, including train/test splitting, overfitting, underfitting, the bias-variance tradeoff, and Linear Regression. These concepts were studied both theoretically and practically using Python and the Scikit-learn library. The project also involved evaluating the model's performance using appropriate metrics and visualizations.

2. Purpose

The main objective of this task was to understand the basic concepts of Machine Learning and apply them by building a Linear Regression model. The task aimed to provide practical experience in preparing data, training a model, evaluating its performance, and understanding how different techniques help improve model reliability and generalization.

3. Objectives

- To understand the purpose of train/test splitting.
- To learn the concepts of overfitting and underfitting.
- To study the bias-variance tradeoff.
- To implement a Linear Regression model using Scikit-learn.
- To evaluate the model using Mean Squared Error (MSE) and R^2 Score.
- To perform cross-validation to assess model performance.
- To compare different train/test split ratios.
- To plot and analyze a learning curve.
- To implement Polynomial Regression and compare its performance with Linear Regression.

4. Tools & Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

5. Methodology

The California Housing dataset was loaded using the Scikit-learn library and converted into a Pandas DataFrame for analysis. The dataset was explored, and the features and target variable were identified before splitting the data into training and testing sets.

A Linear Regression model was then trained using the training data and evaluated on the testing data using Mean Squared Error (MSE) and the R^2 Score. To further assess the model's reliability, different train/test split ratios and five-fold cross-validation were applied. A learning curve was generated to analyze how the model's performance changed with increasing training data. Finally, Polynomial Regression was implemented and compared with the Linear Regression model to observe the effect of increasing model complexity.

6. Results

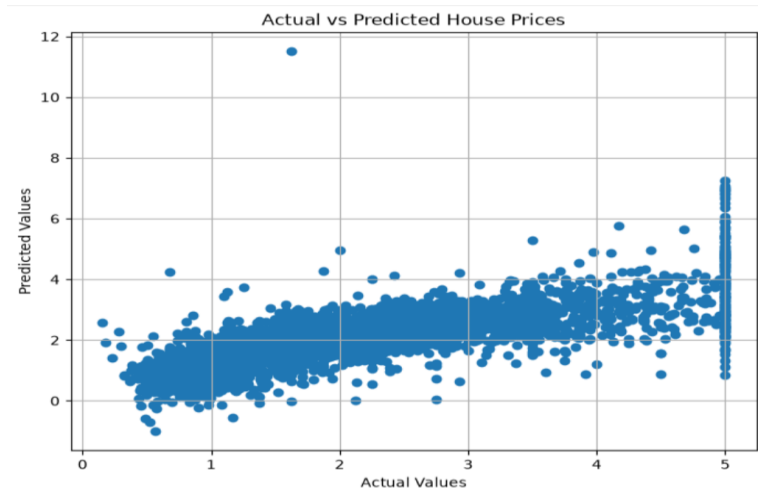
The Linear Regression model achieved satisfactory predictive performance, producing reasonable Mean Squared Error (MSE) and R^2 Score values. The comparison of different train/test split ratios showed consistent results, indicating that the model generalized well across different data partitions. Cross-validation further confirmed the model's stability and reliability.

The learning curve demonstrated how the model's training and validation performance changed as the amount of training data increased, helping identify whether the model was overfitting or underfitting. Additionally, Polynomial Regression was implemented and compared with the Linear Regression model to evaluate the impact of increased model complexity on prediction accuracy.

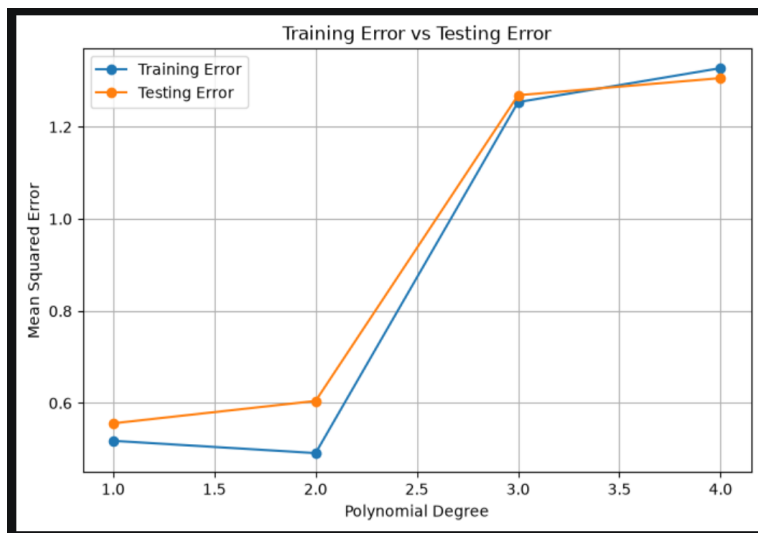
Linear Regression Evaluation Metrics

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Mean Squared Error (MSE): 0.5558915986952442
R2 Score: 0.575787706032451
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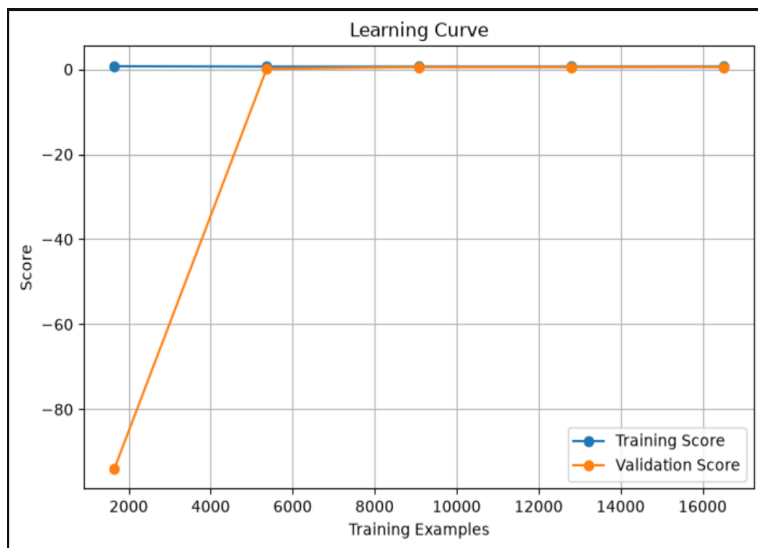
Actual vs Predicted Values



Bias-Variance Tradeoff Graph



Learning Curve



Polynomial Regression Comparison

	Model	MSE	R ² Score
0	Linear Regression	0.559002	0.580835
1	Polynomial Regression	0.604538	0.538664

7. Conclusion

This task provided hands-on experience with the complete machine learning workflow, from data preparation and preprocessing to model training, evaluation, and validation. Linear Regression was successfully implemented to predict house prices using multiple input features, and its performance was evaluated using standard regression metrics.

The comparison with Polynomial Regression demonstrated that increasing model complexity can improve prediction performance but may also increase the risk of overfitting. Overall, this project strengthened my understanding of fundamental machine learning concepts and enhanced my practical skills in building and evaluating predictive models using Python and Scikit-learn.

GitHub Repository Link: <https://github.com/AyeshaBhattiML/PKCERT-Task-07-ML-Foundations>